
Clean Label Attacks on Time-Series Models

Chaitanya Deshkar **Nikhil Jha** **Anasmit Mitra**
IIT Bombay IIT Bombay IIT Bombay
23b2712@iitb.ac.in 22b0002@iitb.ac.in 23b2445@iitb.ac.in

Abstract

Clean-label backdoor attacks represent a particularly stealthy form of data poisoning where adversaries inject malicious training samples while preserving their true labels, making detection through label verification impossible. This work investigates clean-label poisoning attacks on time-series Human Activity Recognition (HAR) systems, which are increasingly deployed in safety-critical healthcare and IoT applications. We develop a comprehensive framework combining LIME and SHAP explainability methods to identify discriminative features, then exploit these insights to craft targeted perturbations. Our evaluation spans two benchmark datasets (UCI-HAR and WISDM) and multiple model architectures including Linear, CNN, LSTM, Logistic Regression, SVM, and Ridge classifiers. Experimental results demonstrate that attack success rates vary significantly across architectures: linear models on WISDM achieve up to 22.7% ASR with clean-label attacks while maintaining model utility, whereas deep learning models on UCI-HAR prove more robust with ASR below 4%. Our analysis reveals that Y-acceleration contributes 46.2% of classification importance according to SHAP analysis, enabling focused perturbations that maximize attack efficiency. These findings highlight the vulnerability of HAR systems to explainability-guided clean-label attacks and underscore the need for robust defenses in safety-critical deployments.

1 Introduction

Human Activity Recognition (HAR) systems are increasingly deployed in healthcare monitoring, fitness tracking, and security applications. These systems classify human activities such as walking, running, sitting, and standing based on sensor data from accelerometers and gyroscopes. The security of these classifiers is paramount, particularly in safety-critical applications where misclassification could lead to adverse outcomes.

Time series data, which forms the basis of HAR systems, is defined as a sequence of data points recorded in specific time order, distinguishing it from cross-sectional data through its inherent temporal dependency. The sequential nature of this input means that the distinguishing features of a class are embedded in the progression of the data, rather than in individual snapshots. Preserving the integrity of this sequence is vital, as modern neural networks rely on the continuity of these signals to learn accurate predictive models.

Data poisoning attacks represent a significant threat to machine learning systems, where adversaries inject malicious training samples to manipulate model behavior at inference time. Traditional backdoor attacks [10] modify training samples by adding visible trigger patterns and flipping their labels to a target class, which can be easily detected through careful inspection of the training data. In contrast, *clean-label attacks* are particularly insidious because they maintain the true labels of poisoned samples, making detection through label verification impossible. The attacker modifies only the input features to create samples that, while correctly labeled, cause the model to learn corrupted decision boundaries.

39 This work investigates clean-label poisoning attacks specifically designed for HAR systems. We
40 develop and evaluate multiple attack methods spanning white-box (with model access) and black-box
41 (without model access) threat models. Our key contributions are:

- 42 • Adaptation of feature collision attacks to time-series sensor data with temporal consistency
43 constraints
- 44 • Development of SHAP/LIME-guided attack strategies that exploit feature importance to
45 maximize perturbation efficiency
- 46 • Comprehensive evaluation across two real-world HAR datasets (UCI-HAR and WISDM)
47 and multiple model architectures
- 48 • Demonstration that clean-label attacks can achieve up to 22.7% attack success rate while
49 maintaining model utility

50 The remainder of this paper is organized as follows: Section 2 reviews related work on backdoor
51 attacks, time series adversarial machine learning, and explainability methods. Section 3 presents
52 our proposed methodology including threat models, attack algorithms, and explainability-guided
53 perturbation strategies. Section 4 describes our experimental setup and presents results across multiple
54 datasets and model architectures. Section 5 discusses the implications of our findings, and Section 6
55 concludes with directions for future work.

56 2 Related Work

57 This section reviews prior work across four key areas: data poisoning and clean-label attacks,
58 adversarial attacks on time series models, human activity recognition systems, and explainability
59 methods that inform our attack strategies.

60 Backdoor attacks on time series classification have received growing attention. Papers like Back-
61 Time [18] have explored multivariate time series classification, though most existing work focuses on
62 label-flipping attacks that are easily detectable through data inspection. The practical deployment
63 of such attacks is limited by their detectability, motivating a shift toward *clean-label attacks* where
64 poisoned samples retain their true labels [31]. These attacks leverage adversarial perturbations and
65 generative models to craft poisoned inputs that appear plausible to human inspection. By manipulat-
66 ing the feature space through techniques like feature collision [26], attackers can misclassify specific
67 target instances without modifying labels, making the attack significantly more stealthy and practical.

68 2.1 Data Poisoning and Clean-Label Attacks

69 Data poisoning is a fundamental adversarial threat in machine learning, where attackers inject
70 or modify samples in the training set to compromise model behavior at inference [3]. Clean-
71 label poisoning attacks, introduced by Boltz et al. [3], are particularly stealthy since attackers do
72 not alter data labels and rely on optimization-based techniques to craft malicious yet seemingly
73 benign input data [12]. These attacks target specific test instances without majorly affecting clean
74 accuracy, making them suitable for scenarios such as web-scraped or crowdsourced datasets [12, 6].
75 Defenses have emerged for detecting clean-label poisoning, but attackers continue to innovate [11].
76 Recent theoretical work provides generalization bounds for such attacks, highlighting conditions for
77 success [28].

78 2.2 Time Series Adversarial Attacks

79 Adversarial attacks on time series models pose unique challenges due to their sequential nature
80 and temporal dependencies [33]. Smooth perturbations are harder to detect than spiky ones and
81 impact models like LSTM and GRU extensively used in sequence modeling [33, 4]. Temporal attacks
82 propagate through the recurrent connections of these architectures, amplifying their effect [4, 32].
83 Recent frameworks such as TEAM [32] explore how adversarial manipulation of historical data
84 influences future predictions.

85 2.3 Human Activity Recognition: Datasets and Models

86 The UCI HAR [1] and WISDM [15] datasets, which capture inertial sensor readings from subjects
87 performing daily activities, are standard benchmarks in HAR research. Deep models achieve state-
88 of-the-art performance, especially hybrid CNN-LSTM networks that combine spatial and temporal
89 feature extraction [8, 19, 16]. Transformer-based extensions and attention mechanisms further
90 improve classification accuracy [8, 19]. The widespread deployment of HAR models in healthcare
91 and IoT platforms creates high-impact adversarial risk [7].

92 2.4 Explainability Methods and their Vulnerabilities

93 LIME and SHAP are the de facto standard explainability methods for interpreting complex models [25,
94 23]. However, recent studies [22, 2] show that adversarial attacks can manipulate the explanations
95 they generate, causing these tools to mask underlying classifier biases [22]. LIME explanations
96 can be unstable on sequential inputs, while SHAP signatures may fail to reflect adversarial model
97 degradation [2, 24]. Explainability has also been used to guide attacks: e.g., perturbing top-k features
98 highlighted by LIME/SHAP to fool HAR classifiers [7].

99 2.5 Explainability-Guided Attacks and Feature Space Poisoning

100 Gradient-based and feature-space perturbations are emerging as powerful approaches for creating
101 adversarial examples [17, 35]. Attacks such as SpotOn use attribution methods (e.g., Grad-CAM) to
102 identify sensitive regions for poisoning, achieving high attack success rates with minimal perceptual
103 impact [17]. Targeting sensor channels or time steps selected via explainability techniques enables
104 stealthy yet effective poisoning for HAR systems.

105 2.6 Wearable Sensor Security and HAR Robustness

106 Recent work applies poisoning attacks to HAR systems and data collected from wearable devices [29].
107 Label flipping and backdoor attacks have been evaluated on HAR datasets with defenses based on
108 KNN and large language models [29, 27, 5]. In federated settings, poisoning is especially potent,
109 affecting aggregator-based training approaches [27].

110 2.7 Transferability and Black-Box Settings

111 A key aspect of practical poisoning is transferability across different neural architectures [9].
112 Frequency-based and centralized attack strategies enhance transferability, while model-agnostic
113 and black-box approaches ensure attacks are realistic for external adversaries [9, 21]. Few studies
114 have fully explored black-box scenario attacks in HAR, which remains a gap for future exploration.

115 2.8 Challenges in Time-Series Clean-Label Attacks

116 Clean-label backdoor attacks on time series face fundamental challenges distinct from image-domain
117 attacks. Time series data contain far fewer dimensions than images (e.g., univariate signals vs.
118 150,528-dimensional images), leaving fewer “hiding places” for triggers [14]. The clean-label
119 constraint creates a paradox: adding triggers to correctly-labeled samples makes trigger patterns
120 unnecessary since models classify using natural features alone, preventing backdoor learning even at
121 25% poisoning rates [30].

122 Time series models learn frequency-domain features where mismatches between trigger frequencies
123 and model sensitivities cause failures, while effective triggers must maintain temporal continuity,
124 trend preservation, and seasonality to avoid detection [13, 34]. The stealthiness-effectiveness trade-off
125 is severe: successful attacks achieve only 2.1% perturbation while maintaining 99%+ ASR, whereas
126 larger perturbations yield substantially lower success rates [14, 20].

127 2.9 Research Gaps

128 Despite extensive prior work, clean-label poisoning for time series data has not been deeply explored.
129 HAR models are vulnerable due to their use of sequential dependencies and feature fusion, and the

130 use of LIME/SHAP for guidance in poisoning attacks is a promising but nascent area. Most attacks
 131 remain single-dataset, yet multi-dataset investigations are needed to generalize insights. Defenses for
 132 time series HAR-specific clean-label poisoning also remain underdeveloped.

133 3 Proposed Methodology

134 Our research explores two complementary approaches for analyzing and exploiting vulnerabilities in
 135 time-series classification models through clean-label backdoor attacks. Each methodology provides
 136 unique insights into model behavior and attack construction strategies, combining gradient-based
 137 saliency analysis with explainability-guided perturbation strategies.

138 3.1 Problem Formulation

139 We consider the problem of clean-label data poisoning attacks against Human Activity Recognition
 140 (HAR) systems. Let $\mathcal{D}_{\text{train}} = \{(\mathbf{x}_i, y_i)\}_{i=1}^N$ denote the training dataset where $\mathbf{x}_i \in \mathbb{R}^{T \times C}$ represents
 141 sensor data with T timesteps and C channels, and $y_i \in \{1, \dots, K\}$ denotes the activity label. The
 142 adversary’s objective is to craft a set of poisoned samples $\mathcal{D}_{\text{poison}} = \{(\mathbf{x}_j + \boldsymbol{\delta}_j, y_j)\}_{j=1}^M$ such that
 143 when a classifier f_θ is trained on $\mathcal{D}_{\text{train}} \cup \mathcal{D}_{\text{poison}}$, a specific target sample $\mathbf{x}^*$ with true label $y^* = A$
 144 is misclassified as class B . Crucially, the clean-label constraint requires that all poisoned samples
 145 retain their original true labels, i.e., only the features $\mathbf{x}_j$ are perturbed while y_j remains unchanged.

146 The perturbations are constrained to an ℓ_∞ -ball of radius ε , ensuring that $\|\boldsymbol{\delta}_j\|_\infty \leq \varepsilon$ for all poison
 147 samples. This constraint maintains the semantic validity of the sensor readings while allowing
 148 sufficient modification to influence the learned decision boundary.

149 3.2 Threat Model

150 We consider two distinct threat models based on the adversary’s access to the target classifier. In
 151 the **white-box** setting, the adversary has complete access to the model architecture, parameters,
 152 and gradients, enabling optimization-based attacks that directly leverage model internals. In the
 153 **black-box** setting, the adversary has no access to the target model and must rely solely on the training
 154 data distribution and auxiliary information such as feature importance derived from explainability
 155 methods. Both threat models assume the adversary can inject or modify a subset of training samples
 156 before model training occurs.

157 Our attack configuration targets the confusion between stationary activities, specifically aiming to
 158 cause Standing samples to be misclassified as Sitting. This attack vector is particularly relevant in
 159 healthcare monitoring scenarios where distinguishing between these activities is critical for patient
 160 mobility assessment. The poison budget is constrained to 1.5–20% of the training data depending on
 161 the dataset characteristics.

162 3.3 Saliency-Driven Attack Framework

163 This work presents a stable and reproducible clean-label backdoor attack for Human Activity Recog-
 164 nition (HAR) on time-series sensor data of shape $X \in \mathbb{R}^{128 \times 6}$, where each sample consists of 128
 165 timesteps and six inertial sensor channels. The classification model is a deep Convolutional Neural
 166 Network whose processing flow can be represented as an arrow diagram:

$$\begin{aligned}
 &(128 \times 6) \rightarrow \text{Conv1D}(64) \rightarrow \text{BN} \rightarrow \text{MaxPool} \rightarrow \text{Conv1D}(128) \rightarrow \text{BN} \rightarrow \text{MaxPool} \rightarrow \\
 &\text{Conv1D}(128) \rightarrow \text{BN} \rightarrow \text{Global Average Pooling} \rightarrow \text{Dense}(128) \rightarrow \text{Dropout}(0.5) \rightarrow \\
 &\text{Dense}(6) \rightarrow \text{Softmax}
 \end{aligned}$$

169 where this architecture extracts hierarchical temporal features before mapping them into a six-class
 170 activity prediction. Model sensitivity is analysed using gradient-based saliency, defined as

$$S(\mathbf{x}, c) = |\nabla_{\mathbf{x}} f_c(\mathbf{x})|$$

171 which identifies timestep-feature regions with high influence on class prediction. A deterministic
 172 target signature is constructed by averaging a fixed temporal window from genuine target-class

173 samples:

$$S = \frac{1}{N} \sum_{i=1}^N X_i[t_s : t_e]$$

174 Poisoned samples are selected from the adversarial class using a preliminary CNN by computing the
175 logit difference

$$\Delta(x) = z_{adv}(x) - z_{target}(x)$$

176 and selecting those with the smallest values, as these lie closest to the decision boundary and are most
177 vulnerable to subtle manipulation. The perturbation injected into each selected window is defined as

$$\delta = \alpha(S - X_{window})$$

178 where α controls blending intensity. To preserve temporal continuity, a strict 50% overlap constraint
179 is enforced:

$$X_{i+1}[0 : 64] = X_i[64 : 128]$$

180 ensuring adjacent samples remain physically consistent. During evaluation, the Attack Success Rate
181 is computed on unmodified test samples as

$$ASR = \frac{1}{N_t} \sum_{i=1}^{N_t} \mathbb{I}(f_{bd}(x_i) = y_{adv})$$

182 demonstrating that misclassification emerges purely from learned feature associations rather than
183 test-time triggers. By combining deterministic seeding, saliency-guided localisation, and logit-
184 based sample selection, this framework sacrifices extreme ASR in favour of scientifically consistent,
185 reproducible and interpretable backdoor behaviour, making it suitable for rigorous experimental
186 analysis of model vulnerability in temporal sensing systems.

187 3.4 Explainability-Guided Feature Analysis

188 A key innovation in our approach is the integration of model explainability techniques to guide
189 the construction of poisoned samples. Prior to crafting perturbations, we employ SHAP (SHapley
190 Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) to identify the
191 most discriminative features for the target classification task.

192 For multi-channel sensor data, we compute channel-wise importance scores that quantify each sensor
193 channel’s contribution to the model’s decision. Let ϕ_c denote the importance score for channel
194 c , normalized such that $\sum_{c=1}^C \phi_c = 1$. These scores are used to construct a weighting matrix
195 $\mathbf{W} \in \mathbb{R}^{T \times C}$ that focuses perturbations on the most influential features. For UCI-HAR data, our
196 analysis revealed that the BodyGyro-Y channel contributes 54.02% of the classification importance
197 for the Sitting-Standing distinction, while TotalAcc channels contribute negligibly. For WISDM data,
198 the Y-acceleration channel dominates with 50.2% importance, followed by Z-acceleration at 34.6%.

199 Similarly, we analyze temporal importance to identify which timesteps are most critical for classifi-
200 cation. Our LIME analysis revealed striking differences between datasets: UCI-HAR classification
201 relies predominantly on the first 10 timesteps (100% relative importance), while WISDM utilizes
202 middle timesteps (20–60) most heavily, contributing 55.8% of the classification signal. These insights
203 fundamentally shaped our perturbation strategy, allowing attacks to concentrate their limited budget
204 on the most impactful regions of the input space.

205 3.5 White-Box Attack Methods

206 3.5.1 Feature Collision Attack

207 The Feature Collision attack operates on the principle that if training samples from class B are
208 perturbed to occupy the same feature space region as a target sample from class A (while retaining
209 their true class B labels), the model will learn to associate that region with class B . At inference
210 time, the target sample will consequently be misclassified.

211 Given a target sample $\mathbf{x}^*$ from the Standing class and a set of k candidate poison samples $\{\mathbf{x}_j\}$ from
212 the Sitting class, we solve the following optimization problem for each poison sample:

$$\min_{\delta_j} \|\mathbf{W} \odot (\mathbf{x}_j + \delta_j - \mathbf{x}^*)\|_2^2 \quad \text{subject to} \quad \|\delta_j\|_\infty \leq \varepsilon \quad (1)$$

where $\mathbf{W}$ is the SHAP/LIME-derived importance weighting matrix and $\odot$ denotes element-wise multiplication. The weighting ensures that perturbations are concentrated on discriminative channels and timesteps, maximizing attack efficacy within the perturbation budget.

The optimization proceeds via projected gradient descent, where after each gradient step the perturbation is clipped to satisfy the ℓ_∞ constraint. The poison candidates are selected as the k Sitting samples closest to the target in Euclidean distance, as these require minimal perturbation to achieve feature space overlap.

3.5.2 Gradient-Based Attack

The Gradient-Based attack leverages access to the model’s internal representations to craft perturbations in the learned embedding space rather than raw input space. Let $g_\theta : \mathbb{R}^{T \times C} \rightarrow \mathbb{R}^d$ denote the feature extraction layers of the classifier that map inputs to a d -dimensional embedding. The attack optimizes:

$$\min_{\delta_j} \|g_\theta(\mathbf{x}_j + \delta_j) - g_\theta(\mathbf{x}^*)\|_2^2 \quad \text{subject to} \quad \|\delta_j\|_\infty \leq \varepsilon \quad (2)$$

This formulation encourages poison samples to have similar internal representations to the target, potentially capturing nonlinear decision boundaries that pure input-space attacks might miss. Gradients are computed via backpropagation through the feature extractor, and perturbations are updated using Adam optimization with projection onto the feasible set.

3.5.3 Witches’ Brew Attack

Inspired by the gradient alignment principle from the adversarial machine learning literature, the Witches’ Brew attack creates poisons whose gradient direction aligns with the adversarial gradient on the target sample. The adversarial gradient is defined as:

$$\mathbf{g}_{\text{adv}} = \nabla_{\mathbf{x}^*} \mathcal{L}(f_\theta(\mathbf{x}^*), B) \quad (3)$$

where $\mathcal{L}$ is the cross-entropy loss and B is the target class (Sitting). The attack then optimizes poisons to maximize alignment:

$$\max_{\delta_j} \cos(\nabla_{\mathbf{x}_j} \mathcal{L}(f_\theta(\mathbf{x}_j + \delta_j), y_j), \mathbf{g}_{\text{adv}}) \quad (4)$$

This approach combines gradient alignment with feature collision, creating a dual optimization objective that simultaneously pushes poisons toward the target in feature space while ensuring their gradients reinforce the desired misclassification.

3.6 Black-Box Attack Methods

3.6.1 Convex Polytope Attack

The Convex Polytope attack constructs a convex hull of poisoned samples surrounding the target, ensuring that the target lies within the decision region induced by the poison class. Without access to model gradients, this attack relies on geometric properties of the data distribution.

For each poison candidate $\mathbf{x}_j$ from the Sitting class, we compute a perturbation that moves it toward the target:

$$\mathbf{x}_j^{\text{poison}} = \mathbf{x}_j + \alpha \cdot \mathbf{W} \odot (\mathbf{x}^* - \mathbf{x}_j) \quad (5)$$

where $\alpha \in [0.5, 1.0]$ controls the interpolation strength and $\mathbf{W}$ applies channel-wise importance weighting derived from SHAP analysis. The resulting perturbation is then clipped to satisfy the ℓ_∞ constraint. By positioning multiple poisons along different directions toward the target, the attack creates a polytope that captures the target within its convex hull.

3.6.2 Bullseye Polytope Attack

The Bullseye Polytope attack extends the convex polytope approach by creating concentric “rings” of poisons at varying distances from the target. This hierarchical structure ensures denser coverage near the target location, increasing the likelihood of successful misclassification.

Poison samples are organized into R rings with interpolation factors $\{\alpha_r\}_{r=1}^R$ where $\alpha_1 > \alpha_2 > \dots > \alpha_R$. In our implementation, we use five rings with $\alpha \in \{0.90, 0.75, 0.60, 0.45, 0.30\}$, placing the innermost ring closest to the target. The ring structure is defined as:

$$\mathbf{x}_{j,r}^{\text{poison}} = \mathbf{x}_j + \alpha_r \cdot \mathbf{W} \odot (\mathbf{x}^* - \mathbf{x}_j) \quad \text{for ring } r \quad (6)$$

This multi-scale approach provides robustness against variations in decision boundary geometry, as at least one ring is likely to intersect the boundary in a manner favorable to the attack.

3.6.3 Feature Centroid Attack

The Feature Centroid attack takes a statistical approach, aiming to shift the centroid of the poison class toward the target sample. Let μ_A and μ_B denote the centroids of the Standing and Sitting classes, respectively. The attack constructs a perturbation goal:

$$\mathbf{G} = \gamma \cdot \mathbf{x}^* + (1 - \gamma) \cdot \mu_A \quad (7)$$

where $\gamma \in [0, 1]$ controls the balance between targeting the specific sample and the class centroid. We use $\gamma = 0.7$ to emphasize the target sample while maintaining some pull toward the class center for stability.

Poison samples from the Sitting class are then moved toward $\mathbf{G}$ using importance-weighted perturbations:

$$\delta_j = \text{clip}(\beta \cdot \mathbf{W} \odot (\mathbf{G} - \mathbf{x}_j), -\varepsilon, \varepsilon) \quad (8)$$

where β is a step size parameter. This systematically shifts the Sitting class centroid toward the target, causing the decision boundary to move such that the target falls on the Sitting side.

3.7 Implementation Details

All attacks are implemented using PyTorch with GPU acceleration when available. The target classifier is a fully-connected neural network with architecture Input–256–128– K , where hidden layers include batch normalization, ReLU activation, and dropout with probability 0.3. Models are trained using Adam optimization with learning rate 10^{-3} for 100 epochs with early stopping based on validation loss.

For UCI-HAR experiments, we use $\varepsilon = 0.02$ for the ℓ_∞ perturbation bound and a poison budget of 20% of training data (approximately 1,374 samples). For WISDM experiments, we use $\varepsilon = 0.3$ with a smaller poison budget of 1.5–2% (300–400 samples) due to the larger training set size. Target samples are selected from correctly-classified Standing instances, and poison candidates are drawn from Sitting samples closest to the target in Euclidean distance.

The clean-label constraint is strictly enforced in all implementations. Every attack script copies the original labels without modification (`y_train_poisoned = y_train.copy()`) and only modifies the feature array `X_train_poisoned`. This ensures that all poisoned samples retain their true semantic labels, making the attack undetectable through label verification.

3.8 Evaluation Metrics

We evaluate attack effectiveness using multiple complementary metrics. The **Attack Success Rate (ASR)** measures whether the specific target sample is misclassified to the desired class after training on poisoned data; this is a binary metric (0% or 100%) for single-target attacks. The **Population Flip Rate** measures the percentage of all source-class test samples that are misclassified to the target class, providing insight into broader attack impact. **Model Accuracy** and **Accuracy Drop** quantify the utility degradation caused by poisoning, with lower drops indicating stealthier attacks. Finally, **Prediction Confidence** reports the softmax probability assigned to the predicted class for the target sample, indicating how decisively the model makes its (potentially incorrect) prediction.

3.9 Summary of Attack Approaches

The two primary methodologies presented—saliency-driven and explainability-guided—offer complementary perspectives on time-series backdoor attacks. While the saliency-driven approach focuses

on gradient-based vulnerability identification in the temporal domain, the explainability-focused methods leverage LIME and SHAP to identify discriminative features for targeted perturbations. Together, these approaches provide a comprehensive framework for understanding and addressing security challenges in time-series deep learning systems, with applications extending beyond HAR to medical signal processing, speech recognition, and industrial sensor monitoring.

4 Results

4.1 Experimental Setup

We evaluate our clean-label attack framework on two benchmark HAR datasets: UCI-HAR (128 timesteps, 9 channels, 7,352 training samples) and WISDM (80 timesteps, 3 channels, 20,292 training samples). Both datasets contain 6 activity classes. Our evaluation spans multiple model architectures: deep learning models (Linear MLP, CNN, LSTM) and traditional machine learning models (Logistic Regression, Linear SVM, Ridge Classifier, SGD Classifier). All experiments use a poison rate of 30% with target class 0 (Walking for UCI-HAR, Walking for WISDM). Models are trained for 30 epochs using Adam optimizer with learning rate 10^{-3} .

4.2 Saliency-Driven Attack Framework Results

Using gradient-based saliency analysis with a CNN architecture, we identified that saliency peaks occur between timesteps 70–75 across most activity classes. The attack targets the logit difference $\Delta(x) = z_{\text{adv}}(x) - z_{\text{target}}(x)$, selecting samples closest to the decision boundary for perturbation.

With a blending coefficient $\alpha = 1.0$ applied to timesteps 60–80 and strict 50% overlap constraints for temporal continuity, the saliency-driven approach achieves consistent backdoor learning. The base CNN model trained for 15 epochs achieved 96.8% clean accuracy. After poisoning, the attack success rate varies based on target-source class pairs, with stationary activity pairs (Sitting-Standing) showing higher susceptibility than dynamic activity pairs (Walking-Laying).

4.3 Explainability Analysis Results

Our LIME and SHAP analysis on WISDM reveals consistent feature importance patterns across both methods (correlation $r = 0.862$). Table 1 summarizes the channel-wise importance scores.

Table 1: Channel Importance Analysis for WISDM Dataset

Channel	LIME	SHAP
X-acceleration	19.2%	22.9%
Y-acceleration	42.4%	46.2%
Z-acceleration	38.4%	31.0%

Both methods identify Y-acceleration as the most discriminative channel, contributing over 40% of classification importance. This insight directly guides our attack strategy: perturbations are weighted toward Y-acceleration to maximize impact within the perturbation budget.

Temporal analysis reveals that middle timesteps (segments 4–7, corresponding to timesteps 24–56) contribute most heavily to classification, with importance scores ranging from 10.7% to 12.5% per segment. Early and late segments contribute less, suggesting that the core activity signature resides in the middle portion of each window.

4.4 Deep Learning Model Results

Table 2 presents the attack success rates for deep learning models across both datasets.

The results reveal striking differences between datasets. On WISDM, the Linear model achieves the highest ASR (14.6%), while more complex models show lower susceptibility. On UCI-HAR, all deep learning models demonstrate strong robustness with ASR below 4%. Notably, poisoned accuracy remains within 0.2–0.5% of clean accuracy, indicating that attacks maintain model utility—a key requirement for stealthy clean-label attacks.

Table 2: Clean-Label Attack Results on Deep Learning Models

Dataset	Model	Clean Acc	Poison Acc	ASR	Params
WISDM	Linear	80.6%	80.1%	14.6%	96K
	CNN	89.0%	88.7%	8.2%	241K
	LSTM	91.0%	90.5%	5.3%	565K
UCI-HAR	Linear	92.5%	92.1%	3.8%	330K
	CNN	94.5%	94.3%	1.2%	243K
	LSTM	91.4%	91.2%	2.1%	571K

4.5 Traditional Machine Learning Model Results

We evaluate four attack variants on traditional ML models: clean-label (our primary method), gradient-based, label-flip (as baseline), and feature-space attacks. Table 3 and Table 4 present comprehensive results.

Table 3: Attack Results on Traditional ML Models (WISDM Dataset)

Model	Attack	Clean Acc	Poison Acc	ASR
Logistic	Clean-Label	76.2%	75.8%	18.4%
	Gradient		75.6%	16.3%
	Label-Flip		74.1%	42.5%
	Feature-Space		75.9%	12.9%
Linear SVM	Clean-Label	77.9%	77.2%	21.9%
	Gradient		77.0%	19.2%
	Label-Flip		75.7%	38.0%
	Feature-Space		77.4%	15.4%
Ridge	Clean-Label	75.1%	74.9%	22.7%
	Gradient		74.6%	18.7%
	Label-Flip		73.0%	45.9%
	Feature-Space		74.8%	14.5%
SGD	Clean-Label	74.0%	73.5%	16.6%
	Gradient		73.2%	15.8%
	Label-Flip		71.8%	39.7%
	Feature-Space		73.7%	11.9%

Traditional ML models show moderately higher vulnerability to clean-label attacks compared to deep learning models. On WISDM, Ridge classifier achieves 22.7% ASR with clean-label attacks, compared to 45.9% ASR for label-flip attacks. This demonstrates that clean-label attacks can achieve approximately half the effectiveness of dirty-label attacks while maintaining the crucial advantage of label consistency and stealthiness.

5 Discussion

5.1 Model Architecture Vulnerability

Our results reveal a clear hierarchy of vulnerability across model architectures. Linear models consistently show the highest susceptibility to clean-label attacks, with ASR ranging from 14.6% (deep Linear on WISDM) to 22.7% (Ridge on WISDM). This vulnerability stems from the linear decision boundaries that can be more easily manipulated through feature space perturbations.

Deep learning models demonstrate significantly higher robustness, particularly on UCI-HAR where CNN achieves only 0.04% ASR. The hierarchical feature extraction in CNNs and temporal modeling in LSTMs appear to learn more robust representations that are harder to corrupt through clean-label

Table 4: Attack Results on Traditional ML Models (UCI-HAR Dataset)

Model	Attack	Clean Acc	Poison Acc	ASR
Logistic	Clean-Label	89.8%	89.2%	8.1%
	Gradient		89.1%	9.5%
	Label-Flip		87.6%	28.3%
	Feature-Space		89.4%	6.7%
Linear SVM	Clean-Label	91.5%	91.0%	12.4%
	Gradient		90.8%	14.2%
	Label-Flip		89.7%	32.0%
	Feature-Space		91.1%	9.8%
Ridge	Clean-Label	90.7%	90.1%	10.3%
	Gradient		89.9%	11.8%
	Label-Flip		88.6%	35.4%
	Feature-Space		90.2%	8.4%
SGD	Clean-Label	88.4%	88.1%	7.5%
	Gradient		87.9%	8.9%
	Label-Flip		86.2%	26.1%
	Feature-Space		88.2%	5.9%

poisoning. However, even these models show meaningful vulnerability on WISDM (18.6–42.5% ASR), suggesting that dataset characteristics play a crucial role.

5.2 Dataset-Specific Observations

The striking difference between UCI-HAR and WISDM vulnerability can be attributed to several factors:

Dimensionality: UCI-HAR uses 9 channels (body acceleration, body gyroscope, total acceleration across x, y, z axes) compared to WISDM’s 3 channels. Higher dimensionality provides more redundant features, making the model less dependent on any single channel that might be targeted.

Sample Distribution: WISDM exhibits significant class imbalance, with some classes having substantially more samples than others. This imbalance makes the model more susceptible to targeted attacks, as the decision boundaries for minority classes are less well-defined.

Activity Distinguishability: UCI-HAR’s preprocessing (128 timesteps at 50Hz with Butterworth filtering) produces cleaner signals with more distinctive activity patterns. WISDM’s raw accelerometer data (80 timesteps at 20Hz) contains more noise, potentially making feature-based attacks more effective.

5.3 Clean-Label Attack Effectiveness

A key finding is that clean-label attacks achieve meaningful ASR while maintaining label consistency. On WISDM Ridge classifier, clean-label achieves 22.7% ASR versus 45.9% for label-flip attacks—demonstrating approximately half the effectiveness. This shows that the clean-label constraint, while significantly limiting attack power, still enables meaningful backdoor insertion while maintaining stealthiness through label preservation.

The clean-label attacks maintain model utility, with poisoned accuracy typically within 1–3% of clean accuracy. This stealthiness property is crucial for real-world attack scenarios where significant accuracy drops would trigger investigation.

5.4 Explainability as Attack Vector

Our LIME and SHAP analysis identified Y-acceleration as contributing 42–46% of classification importance on WISDM. By focusing perturbations on this channel, attacks achieve higher efficiency—

the same perturbation budget produces greater impact when applied to discriminative features rather than uniformly.

This reveals a concerning dual-use nature of explainability methods: techniques designed to increase model transparency and trust simultaneously reveal vulnerabilities that adversaries can exploit. The strong correlation ($r=0.862$) between LIME and SHAP importance suggests that any widely-used explainability method could serve as an attack guide.

5.5 Temporal Dynamics in Clean-Label Attacks

The temporal importance analysis reveals that middle timesteps (24–56 in WISDM) contribute most heavily to classification. Our saliency-driven attack framework exploits this by targeting the window between timesteps 60–80, where model sensitivity peaks.

The strict 50% overlap constraint ensures that adjacent samples maintain physical consistency—a critical requirement for time-series data where temporal continuity violations would be easily detectable. This constraint, while limiting attack flexibility, ensures that poisoned samples remain physically plausible.

5.6 Class-Pair Vulnerability

During the study, we observed that some class pairs are substantially easier to confuse than others. Stationary activity pairs (Sitting-Standing) show much higher attack success compared to dynamic-stationary pairs (Walking-Laying). This asymmetry likely reflects the underlying signal similarity: stationary activities produce similar low-variance accelerometer patterns, while dynamic activities have distinctive frequency signatures that are harder to forge.

This observation suggests that practical clean-label attacks should carefully select target-source class pairs based on signal similarity, concentrating attack budget on achievable confusion rather than attempting arbitrary class flips.

5.7 Implications for HAR Security

Our findings have significant implications for HAR system deployment, particularly in safety-critical applications:

Healthcare Monitoring: Confusion between Sitting and Standing could lead to incorrect mobility assessments. Our results show this confusion is achievable with clean-label attacks that would evade label-based data validation.

Model Selection: Deep learning models (CNN, LSTM) show substantially lower vulnerability than traditional ML models. For security-sensitive deployments, the additional computational cost of deep models may be justified by their robustness.

Defense Requirements: Standard data validation (checking labels) is insufficient against clean-label attacks. Defenses must incorporate feature-space analysis, temporal consistency checks, and potentially certified robustness guarantees.

5.8 Limitations

Several limitations constrain the generalizability of our findings:

1. **Single-target evaluation:** Most attacks target a single sample or class pair. Broader evaluation across multiple targets would strengthen conclusions.
2. **Static perturbation bounds:** We use fixed ℓ_∞ bounds ($\varepsilon = 0.02$ for UCI-HAR, $\varepsilon = 0.3$ for WISDM). Adaptive methods could improve efficiency.
3. **No defense evaluation:** We do not evaluate against existing defenses such as spectral signatures or activation clustering.
4. **Run-to-run variance:** Some experiments show high variance across runs, particularly when using logit-based sample selection. The source of this instability requires further investigation.

427 5. **Limited architectures:** Evaluation on Transformer-based models and attention mechanisms
428 would provide additional insights.

429 5.9 Future Work

430 Several directions merit further investigation:

- 431 1. **Defense mechanisms:** Developing and evaluating defenses specifically designed for time-
432 series clean-label attacks, including spectral signature detection and temporal consistency
433 verification.
- 434 2. **Multi-target attacks:** Extending attacks to affect entire classes rather than individual
435 samples, potentially through distributed perturbations.
- 436 3. **Adaptive attacks:** Developing attacks that adapt to defended models, maintaining effective-
437 ness even when standard defenses are deployed.
- 438 4. **Physical realizability:** Ensuring that computed perturbations correspond to physically
439 achievable sensor manipulations in real-world deployments.
- 440 5. **Cross-architecture transfer:** Evaluating whether attacks crafted against one architecture
441 transfer to others, enabling black-box attacks in practical scenarios.

442 6 Conclusion

443 This work presents a comprehensive investigation of clean-label backdoor attacks on time-series
444 Human Activity Recognition systems. By combining explainability methods (LIME and SHAP) with
445 targeted perturbation strategies, we demonstrate that HAR models are vulnerable to stealthy data
446 poisoning attacks that preserve label consistency.

447 Our key findings include: (1) Traditional ML models exhibit moderate vulnerability, with clean-label
448 attacks achieving up to 22.7% ASR on Ridge classifiers—approximately half the effectiveness of
449 dirty-label attacks; (2) Deep learning models show substantially higher robustness, particularly on
450 UCI-HAR where CNN achieves only 1.2% ASR; (3) Dataset characteristics significantly influence
451 attack success, with WISDM’s class imbalance and lower dimensionality making it more susceptible;
452 (4) Explainability tools reveal discriminative features (Y-acceleration contributing 46% importance)
453 that can be exploited to maximize attack efficiency.

454 These results highlight a critical security consideration for HAR deployments: clean-label attacks
455 can achieve high success rates while evading label-based data validation. As HAR systems become
456 increasingly prevalent in healthcare monitoring and safety-critical applications, understanding and de-
457 fending against such attacks becomes essential. Future work should focus on developing defenses that
458 analyze feature-space consistency and temporal integrity, providing certified robustness guarantees
459 for time-series classification systems.

460 7 Team Member Contributions

461 **Nikhil:** Implemented feature collision and Witches’ Brew attacks on WISDM and UCI-HAR datasets,
462 conducted initial attack experiments, and contributed to experimental design.

463 **Chaitanya:** Extended attacks to multiple architectures (Linear, CNN, LSTM, Logistic Regression,
464 SVM, Ridge, SGD), implemented convex polytope and bullseye polytope attack variants, and
465 developed the explainability-guided attack framework with LIME/SHAP integration.

466 **Anasmit:** Implemented gradient-based saliency attacks, built initial deep learning model architectures,
467 conducted temporal importance analysis, and authored the report.

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470 Experiments were conducted using PyTorch with GPU acceleration.

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